

- 6 (a) (i) to increase flux linkage (with secondary coil) **B1**
- (ii) e.m.f. (induced only) when flux (in core/coil) is changing **B1**
 constant / direct voltage gives constant flux / field **B1**
- (b) (i) $N_S / N_P = V_S / V_P$ **C1**
 $N_S = (52 / 150) \times 1200$
 $= 416$ turns **A1**
- (ii) 0 ms or 7.5 ms or 15.0 ms or 22.5 ms **A1**
- (c) (i) *either*
 mean power = $V^2 / 2R$ and $V = 52$ (V) **C1**
 $R = 52^2 / (2 \times 1.2)$
 $= 1100$ (1127) Ω **A1**
or
 mean power = V^2 / R and $V = 52 / \sqrt{2}$ (= 36.8 V) **C1**
 $R = 36.8^2 / 1.2$
 $= 1100 \Omega$ **A1**
- (ii) sinusoidal shape with troughs at zero power **B1**
 only 3 'cycles' **B1**
 each 'cycle' is 2.4 W high and zero power at correct times **B1**